

MANGROVE MOUNTAIN, Australia

WMO: 957740

Lat: 33.2894S

Lon: 151.2108E

Elev: 1001

StdP: 14.17

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.8	39.8	26.0	20.7	59.5	28.9	23.8	54.6	19.1	57.1	16.5	56.0	3.4	310	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.6	93.2	69.0	88.8	68.3	84.8	67.6	73.8	84.8	72.1	81.8	70.8	79.1	8.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.0	118.6	77.1	69.6	113.0	74.9	68.3	108.1	73.4	38.1	84.9	36.5	81.9	35.3	79.5	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.9	12.1	33.9	101.5	1.4	4.1	32.8	104.5	32.0	106.9	31.2	109.2	30.2	112.2		
(5)			32.5	77.5	1.1	2.0	31.7	78.9	31.1	80.1	30.4	81.2	29.6	82.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.6	71.4	70.5	67.6	62.5	56.5	52.4	51.1	53.0	58.2	62.2	65.7	68.9		
	DBStd	8.74	5.91	5.38	4.94	4.59	4.16	3.57	3.69	4.52	5.71	6.44	6.51	6.05		
	HDD50	65	0	0	0	0	3	16	28	17	1	0	0	0		
	HDD65	2038	9	8	27	101	264	378	429	373	220	134	69	26		
	CDD50	4305	664	574	546	376	204	87	63	109	246	379	470	587		
	CDD65	806	208	162	108	27	1	0	0	1	15	48	88	148		
	CDH74	5839	1579	1005	542	136	5	0	0	2	146	460	773	1191		
	CDH80	2168	690	368	143	23	0	0	0	0	31	140	298	475		
(14) Wind	WSAvg	5.4	6.2	5.7	5.0	4.2	4.7	4.9	5.3	5.6	5.8	5.8	5.9	5.8		
(15) Precipitation	PrecAvg	39.2	4.0	5.6	3.9	3.5	2.9	3.7	1.9	2.5	2.0	2.2	3.3	3.5		
	PrecMax	67.2	13.0	20.5	10.3	14.4	10.5	16.3	6.1	10.4	6.1	7.7	7.7	8.0		
	PrecMin	25.2	0.5	0.6	0.3	0.1	0.4	0.6	0.0	0.0	0.0	0.4	0.2	1.1		
	PrecStd	10.3	2.5	4.6	2.2	3.3	2.4	3.1	1.5	2.5	1.5	1.8	1.9	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.1	96.5	89.2	83.5	74.1	67.2	68.4	72.9	84.1	89.7	95.2	97.1		
		MCWB	71.6	72.2	68.8	64.5	60.6	56.7	55.9	55.1	59.8	61.6	66.6	69.0		
	2%	DB	93.0	89.6	83.8	77.7	70.1	63.5	63.8	68.7	78.4	83.9	87.9	90.6		
		MCWB	70.8	70.8	68.5	63.8	59.1	55.2	52.8	54.2	57.9	61.9	66.5	68.0		
	5%	DB	87.6	84.0	79.8	74.1	67.2	61.4	61.1	65.5	73.4	79.2	82.4	85.2		
		MCWB	69.8	70.3	67.9	63.1	57.8	54.3	52.0	53.2	56.9	60.8	64.5	67.4		
	10%	DB	82.7	79.8	76.1	70.9	64.6	59.4	58.6	62.3	69.3	74.4	77.5	80.5		
		MCWB	69.5	69.2	66.6	62.0	57.0	53.4	51.0	51.7	55.9	60.1	64.5	66.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	76.3	73.6	68.5	64.3	59.6	59.5	59.6	63.3	67.5	72.2	74.0		
		MCDB	89.7	88.5	82.0	76.2	67.6	63.1	64.3	68.2	74.6	77.8	84.9	86.6		
	2%	WB	74.0	73.6	71.1	66.3	62.2	57.6	56.0	57.1	61.2	65.4	69.6	71.4		
		MCDB	86.3	82.5	79.0	72.5	66.0	60.6	59.6	63.1	70.4	75.4	81.3	83.2		
	5%	WB	72.1	71.9	69.6	64.8	60.3	56.4	54.2	55.3	59.6	63.7	67.5	69.7		
		MCDB	82.4	79.4	75.8	70.1	64.7	59.3	57.9	61.9	68.0	73.0	77.4	79.7		
	10%	WB	70.6	70.3	68.2	63.6	58.6	55.0	52.5	53.5	58.0	62.1	65.8	68.2		
		MCDB	78.9	76.9	73.2	68.7	63.0	58.0	56.6	59.8	66.1	70.3	74.1	76.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	16.0	15.5	16.0	16.2	14.1	15.7	18.1	19.2	19.1	18.1	18.5		
		MCDBR	28.7	24.7	22.0	20.5	19.8	17.0	18.9	21.9	25.4	27.4	27.9	27.5		
		MCWBR	10.3	9.8	9.3	9.7	10.5	10.3	10.8	11.1	10.9	10.9	10.3	10.1		
		MCDBR	22.9	19.9	18.5	17.3	15.6	12.9	14.3	17.9	20.1	21.7	22.4	22.7		
	5% WB	MCWBR	9.7	9.2	8.7	9.3	9.3	8.9	9.7	10.5	10.6	10.5	10.2	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.383	0.375	0.344	0.321	0.307	0.298	0.291	0.297	0.330	0.353	0.378	0.375		
	taud		2.460	2.508	2.583	2.621	2.607	2.633	2.639	2.614	2.511	2.470	2.438	2.467		
	Ebn at Noon		303	299	298	288	276	271	279	292	297	302	302	306		
	Edh at Noon		38	35	30	26	23	21	22	25	32	36	38	38		
(44) All-Sky Solar Radiation	RadAvg		2051	1787	1517	1226	971	772	903	1193	1572	1806	1926	2094		
	RadStd		146	156	135	104	72	67	70	72	86	161	179	179		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.72	-1.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		+0.81	N/A	N/A	+1.51	N/A	N/A	-4	-155	+342	+184

Nomenclature: See separate page